

IPR and Role of Scientists.

IPR

Rights given to persons over the creations of their minds

Copyright : Literary and Artistic works, 60 years.

Patent : exclusive right of invention, country specific, product or process, 20 years.

List of Non Patentable items

Compulsory Licensing

Parallel Import

Evergreening of Patent

TKDL

Utility model : less strict than patent, between 6 and 15 years.

Trademark : distinguishes goods or services of an enterprise from another, initially registered for 10 years.

Industrial design : 3d features, 15 years.

Geographical Indications : 10 years, for specific geographically attributable products.

Trade secrets : Remain Valid as long as independently discovered.



IPR and Role of Scientists.

Nobel prize in Medicine / Physiology

- Swedish Geneticist Svante Paabo
- Sequenced genome of neanderthal and used mitochondrial DNA analysis to understand human evolution and movements,
- Use of Mitochondrial DNA as it does not get influenced by only 1 parent.
- Related Term : denisovans

Standard Model Of Particle Physics

- Finding the fundamental particles of the universe.
- Further division of neutron and proton was possible,
- 6 types of quarks, 6 types of leptons, 4 fundamental forces but gravity is not in the standard model.
- Particles responsible for forces : Gluon, Photon, W/Z boson.
- How do these particles get mass? : explained by higgs boson first detected at the large hadron collider.



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Nobel Prize in Chemistry

- Click chemistry and Bioorthogonal reactions.
- Gets past the problem of unwanted products coming up due to artificial copying of natural processes.
- Important : Copper Catalysed azide - alkyne cycloaddition.
- Bioorthogonal chemistry : using this process in living beings without copper ions. This helps us tag glycans.
- Applications : develop pharmaceuticals, herbicides, understand complex biological process, drug development and targeting.

Nobel Prize in Physics

- Quantum Entanglement, Quantum Teleportation, Quantum Entanglement swapping and theoretical work on bell inequalities.
- Related term : no cloning no deleting theorem.
- Entanglement Swapping

